

WINE QUALITY PREDICTION - EDA REPORT

DATASET OVERVIEW

Original Records: 21000

Original Columns: 12

Records After Duplicate Removal: 14940

Input Features: 11

Target Variable: quality

Missing Values: 0

Duplicate Records Found: 6060

Numerical Features: 12

Categorical Features: 0

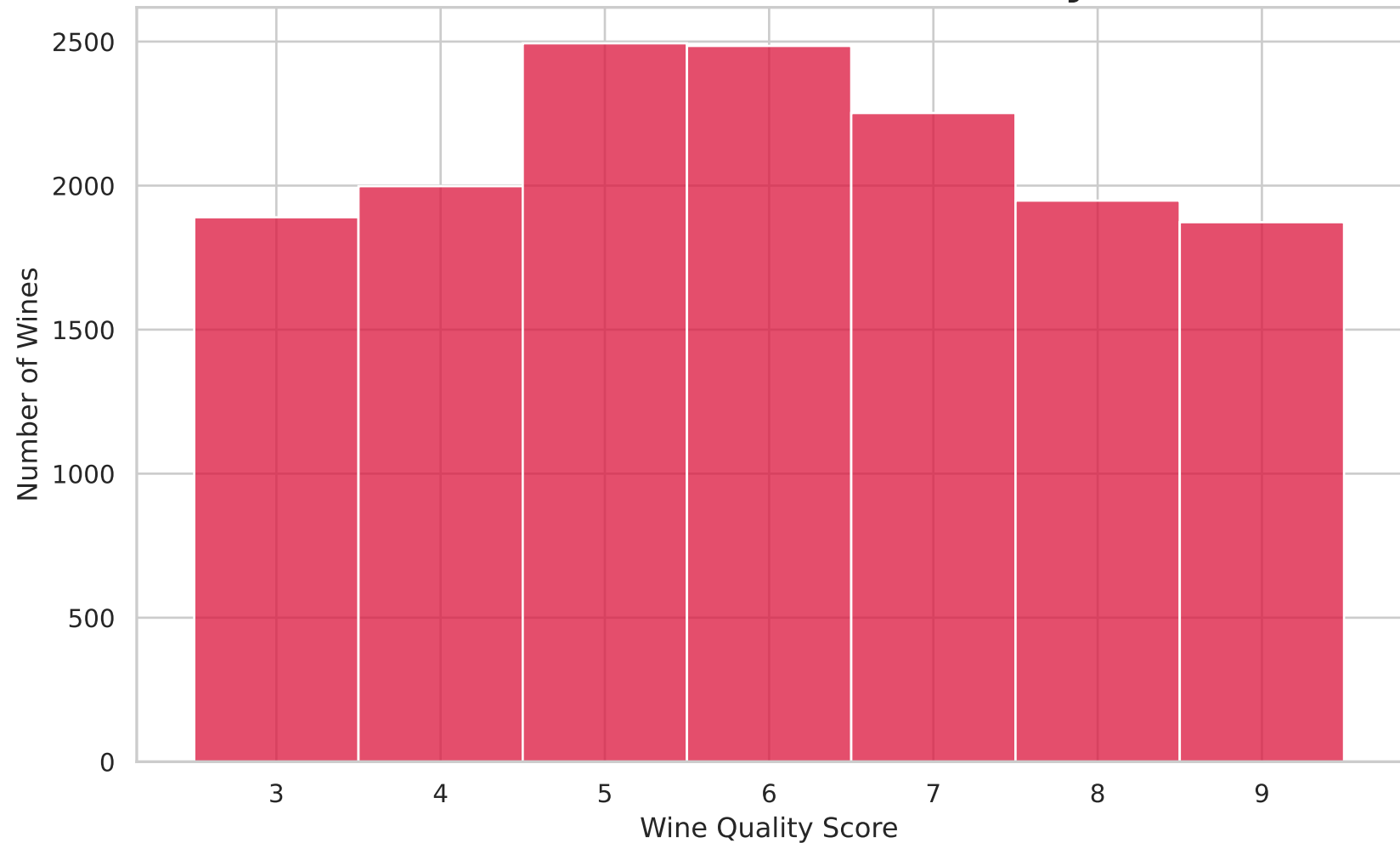
Quality Values:

[np.int64(3), np.int64(4), np.int64(5), np.int64(6), np.int64(7), np.int64(8), np.int64(9)]

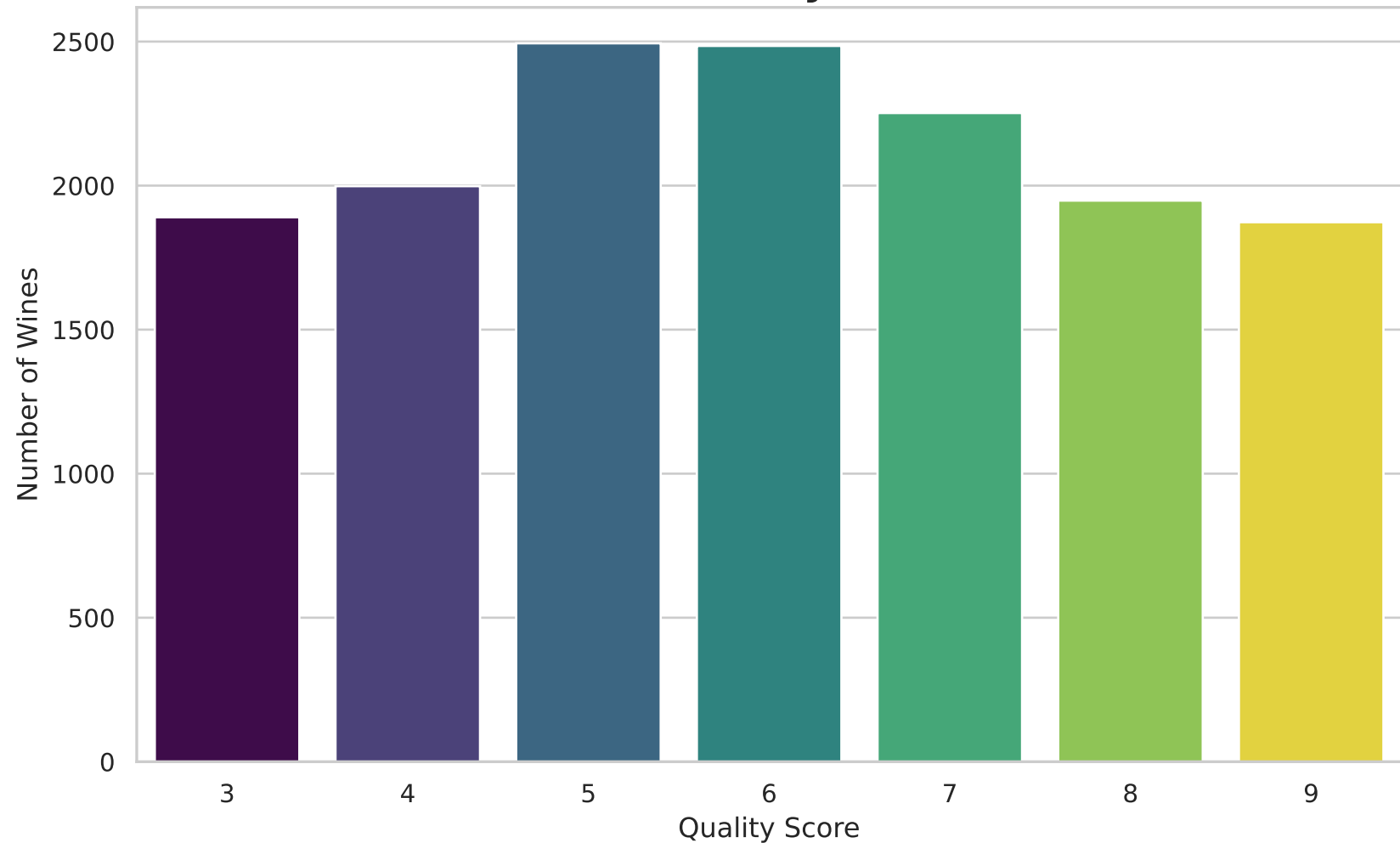
Statistical Summary

	fixed_acidity	volatile_acidity	citric_acid	residual_sugar	chlorides	free_sulfur_dioxide	total_sulfur_dioxide	density	pH	sulphates	alcohol	quality
count	14940.0	14940.0	14940.0	14940.0	14940.0	14940.0	14940.0	14940.0	14940.0	14940.0	14940.0	14940.0
mean	9.62	0.746	0.763	29.422	0.191	122.365	220.595	1.009	3.165	0.989	11.261	5.974
std	2.441	0.37	0.393	19.674	0.127	78.781	102.328	0.012	0.172	0.414	1.184	1.908
min	3.8	0.08	0.0	0.6	0.009	1.0	6.0	0.987	2.72	0.22	8.0	3.0
25%	7.3	0.37	0.37	6.5	0.059	38.0	137.0	0.996	3.04	0.58	10.4	4.0
50%	9.7	0.785	0.82	36.0	0.192	136.55	228.0	1.011	3.15	1.02	11.3	6.0
75%	11.6	1.06	1.1	46.2	0.29	190.9	307.0	1.019	3.28	1.34	12.2	8.0
max	15.9	1.58	1.66	65.8	0.611	289.0	440.0	1.039	4.01	2.0	14.9	9.0

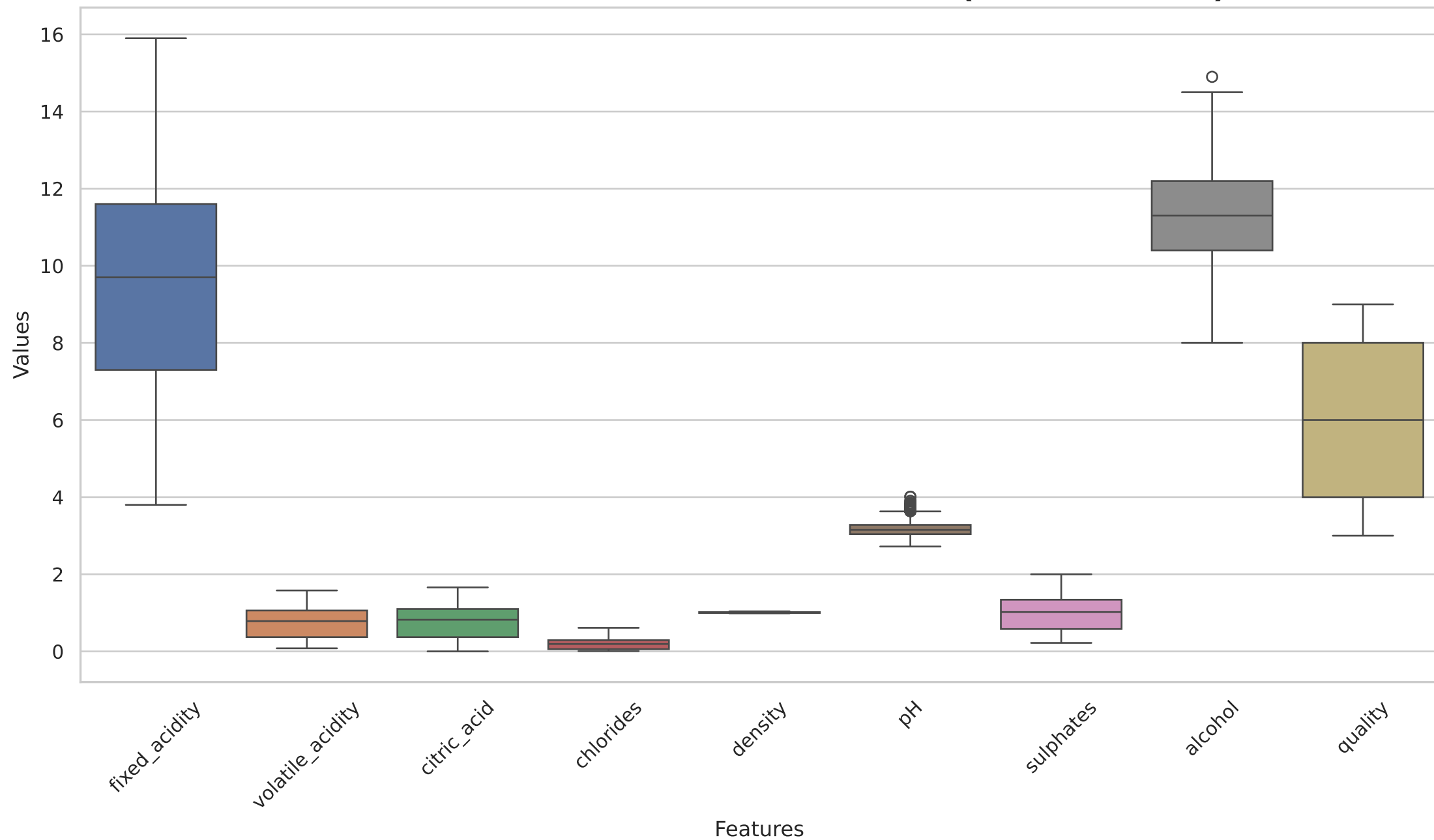
Distribution of Wine Quality



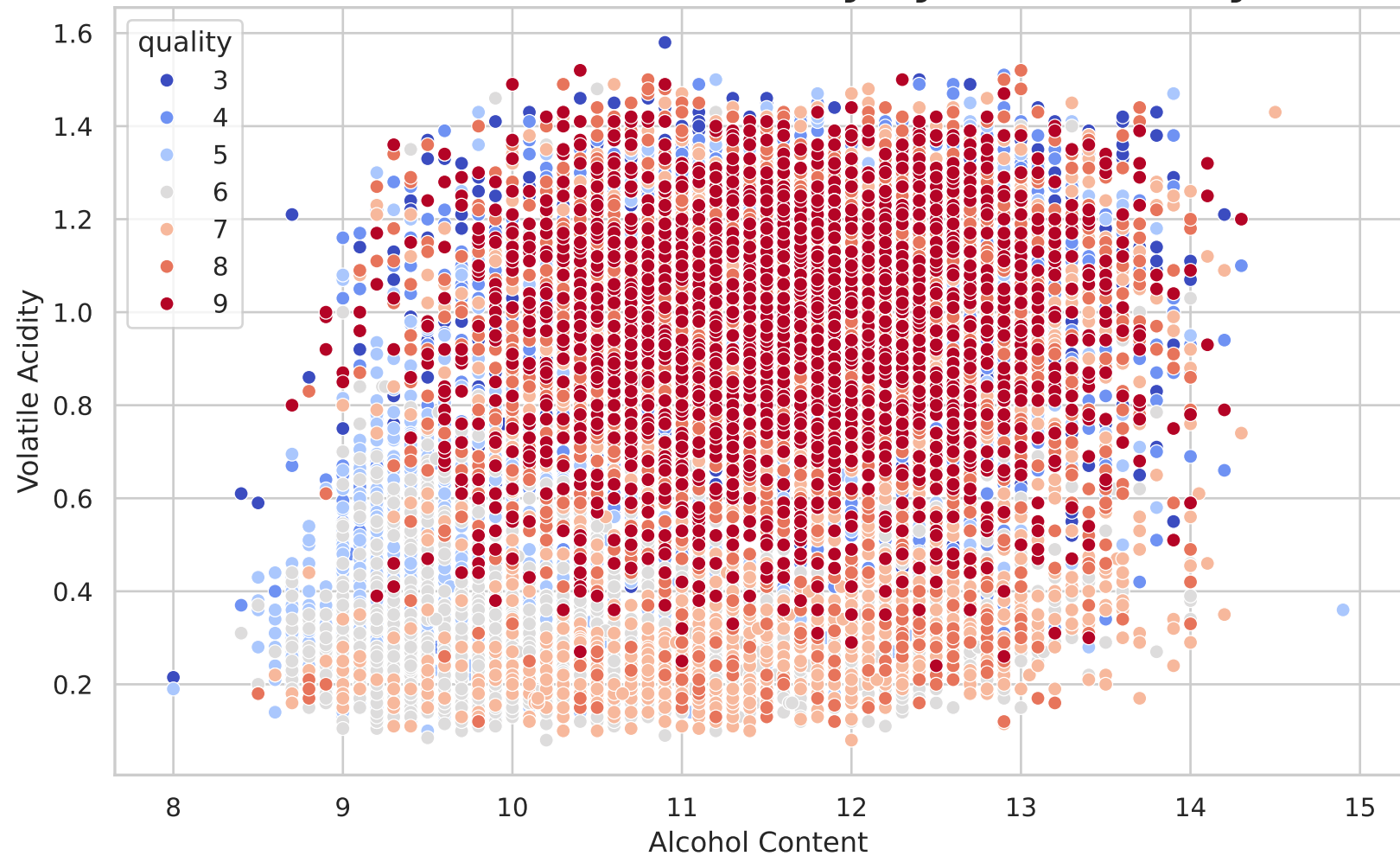
Wine Quality Counts



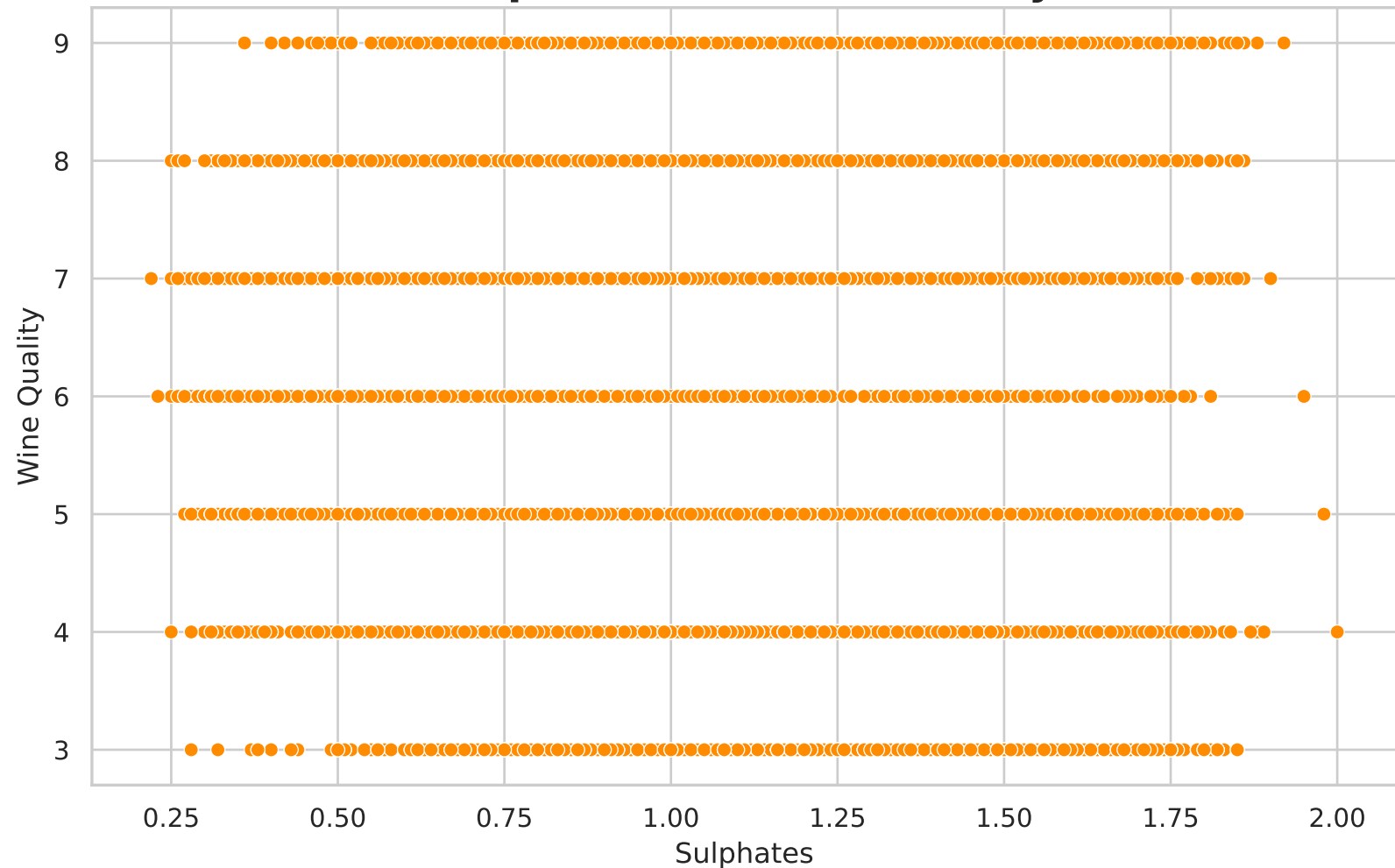
Box Plot of Scaled Chemical Features (Outlier Check)



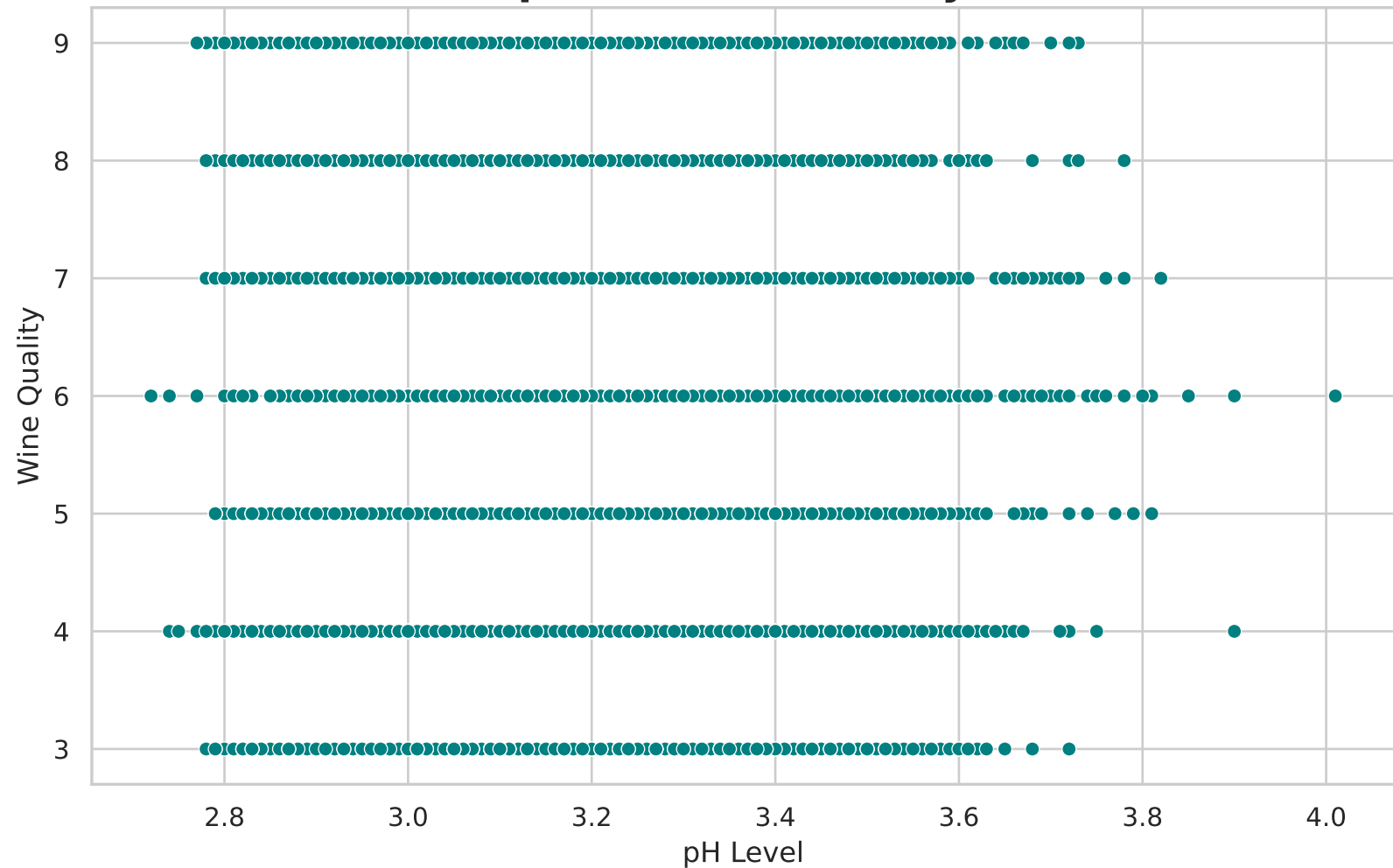
Alcohol vs Volatile Acidity by Wine Quality



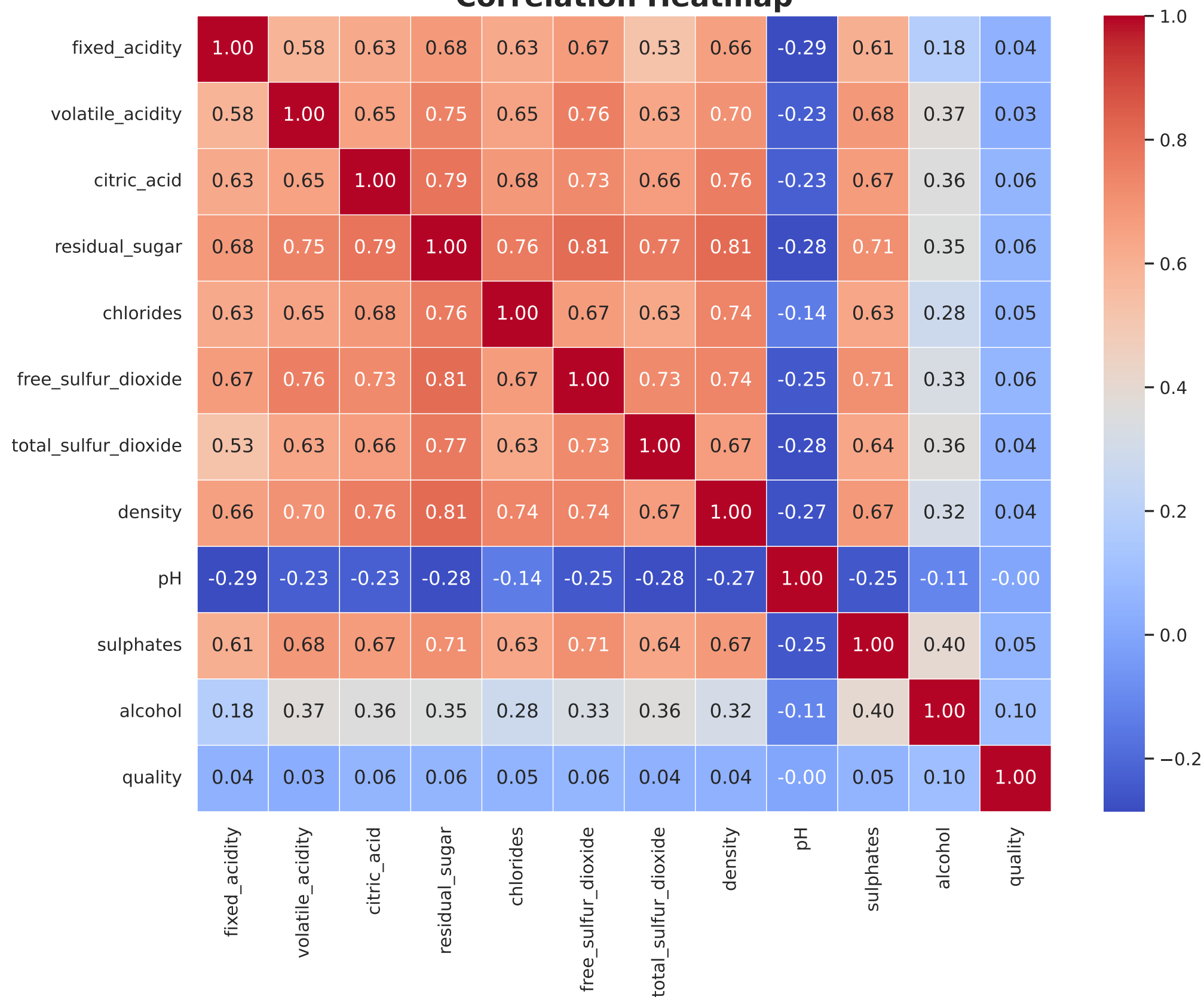
Sulphates vs Wine Quality



pH vs Wine Quality



Correlation Heatmap



EDA Findings and Conclusion

EDA OBSERVATIONS AND CONCLUSION

- The dataset contains 14940 unique wine records and 11 physicochemical features.
- No missing values were found.
- 6060 duplicate records were identified and handled.
- Wine quality scores range from 3 to 9.
- Higher alcohol content and sulphates correlate positively with higher quality ratings.
- High volatile acidity is negatively associated with wine quality.
- Box plots were used to detect outliers across features.

CONCLUSION

Exploratory Data Analysis helped uncover key patterns, feature correlations, and distributions in the Wine Quality dataset, laying a strong foundation for predictive Machine Learning classification models.